

Innovators

“Our Goal Was To Build A PROCESSOR IN INDIA, Without Foreign Dependence.”

India is developing its own processors from the ground up. Through its Vega chips, CDAC aims to deliver full control, AI capabilities, and custom designs, reshaping how industry, startups, and students engage with silicon. Recently, Ashwini Kumar Sinha and Nidhi Agarwal from EFY spoke to Krishnakumar Rao of CDAC about the architecture, challenges, and vision behind building an indigenous IP processor...



KRISHNAKUMAR RAO
DEPUTY DIRECTOR, CDAC

What was the inspiration behind starting an indigenous processor project in India?

Between 2014 and 2016, India's policymakers realised the absence of indigenous processors with full ownership and control. This led to the decision to develop an Indian processor with complete control over the design, source code, and all technical aspects, without dependence on foreign entities. With this intent, discussions began at the government level, followed by a proposal submitted to the Ministry of Electronics and Information Technology (MeitY). After approval, formal project activities began around 2018, although preliminary study and analysis had already been underway.

How much of the design is standard RISC-V, and how much is custom?

The processor is fully custom in terms of microarchitecture, but it is compliant with a specific flavour of the RISC-V instruction set architecture (ISA). That means different designers can build their own processors, each with its own

microarchitecture, while still targeting the same RISC-V ISA standard. Functionally, the processors will execute the same instructions and produce the same results, assuming they are implemented correctly. From a functional perspective, they are equivalent. The difference lies in how each processor is implemented internally. While the ISA remains standard, the underlying microarchitecture is entirely custom to each design.

Does your processor act as a base that allows engineers to design their own applications and add custom elements as needed?

The processor serves as a base; you can add your secret IP or other custom modules and build the final product. We support this for FPGA-based solutions as well. For certain strategic applications with low-volume requirements, we deploy the processor on FPGA boards. In these cases, the board integrates our processor, additional in-house IP, and sometimes customer-provided IP, creating a fully indigenous

solution tailored to their needs.

What software tools, compilers, and programming environments are supported for the VEGA processor, and do developers need separate modules for IoT connectivity like Wi-Fi or Bluetooth?

The VEGA processor is compatible with the **open source compiler toolchain provided by RISC-V**. This means anyone can design, compile, and run programs on VEGA using standard open source tools. Developers can also use the Arduino IDE, which is widely used in academia. The underlying GNU toolchain operates in the background, and a board support package has been created to enable compatibility with Arduino. Although the toolchain currently requires no customisation, future support for custom instructions may require compiler-level modifications. **On the hardware side, VEGA chips do not currently integrate IoT connectivity such as Wi-Fi or Bluetooth. To enable these features, a separate external module must be added.**